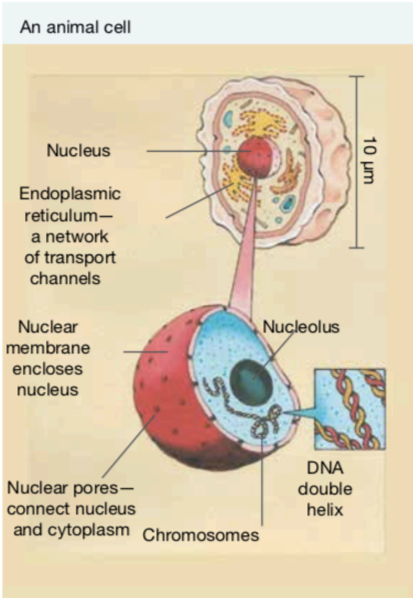
GENETIC INFORMATION

Structure of DNA

The nucleus is the control centre of our cells and it contains all of the genetic information of an individual.

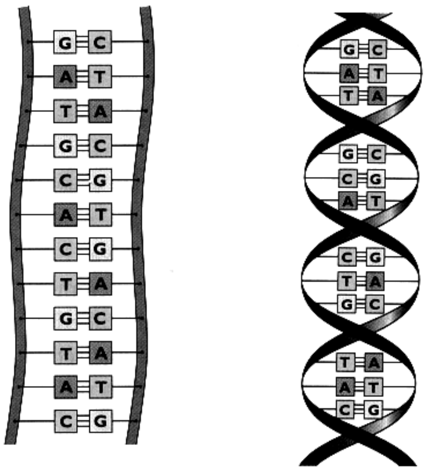


Tightly-coiled, thread-like bodies called ***chromosomes*** contain this information.

Chromosomes are only visible when a cell is undergoing cell division.

Each chromosome is made up of genetic material called ***deoxyribonucleic acid (DNA)***.

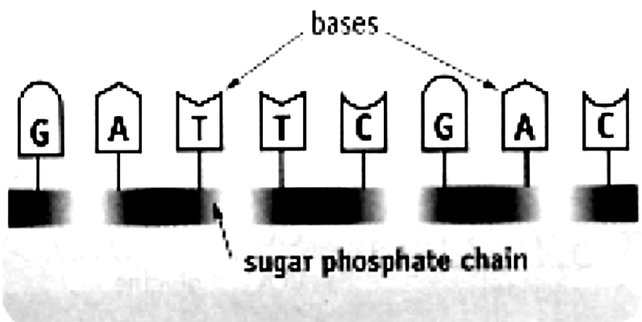
The DNA molecule is a double helix, a twisted ladder with "rungs" made up of four bases - adenine (A), guanine (G), thymine (T) and cytosine (C).



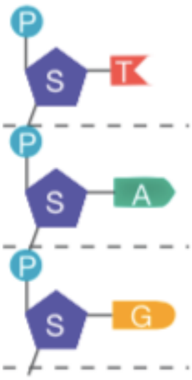
Ladder-like structure twists to form a double helix.

These occur in pairs - A and T, G and C.

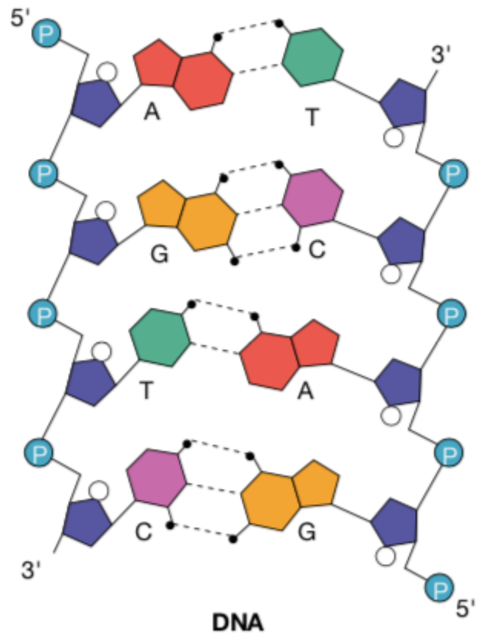
The order of these bases and their combinations determines the ***genes*** found on the chromosome.

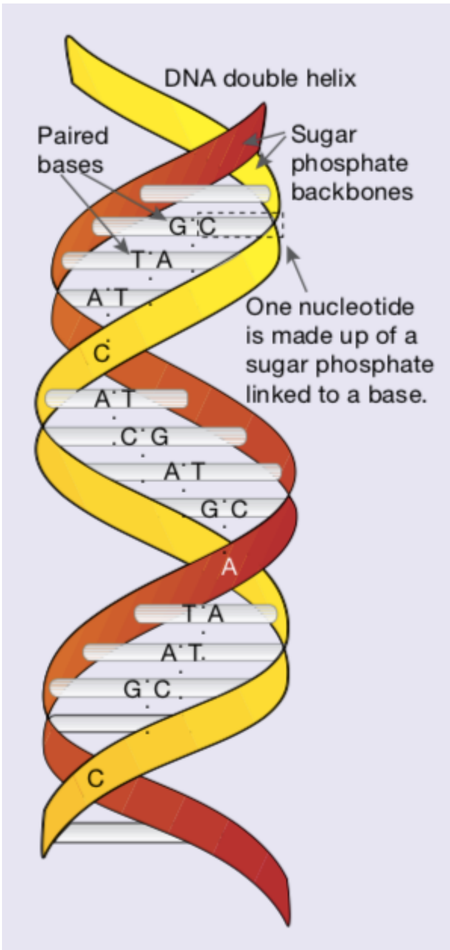


Nitrogenous bases are attached to the sugar part of the nucleotide.



DNA is made up of repeating units called ***nucleotides***. Each of these nucleotides consisted of a sugar, a phosphate group and a nitrogenous base.





Genetic Code

DNA provides the code for cells to produce proteins that are made up of amino acids. The number of amino acids in a protein can vary from 50 to 50, 000 or more. It is the DNA code that determines an organism's characteristics.

The bases along the DNA strand are arranged in ***triplet codes***. Each triplet code represents one type of amino acid.

e.g. AGA is for serine.
CCA is for glycine.

Some amino acids have more than one triplet code.

e.g. ATA, ATG is for tyrosine.

DNA code	amino acid	DNA code	amino acid	DNA code	amino acid	DNA code	amino acid
AAA	phenylalanine	GAA	leucine	TAA	isoleucine	CAA	valine
AAG		GAG		TAG		CAG	
AAT	leucine	GAT		TAT		CAT	
AAC		GAC		TAC	methionine	CAC	
AGA	serine	GGA	proline	TGA	threonine	CGA	alanine
AGG		GGG		TGG		CGG	
AGT		GGT		TGT		CGT	
AGC		GGC		TGC		CGC	
ATA	tyrosine	GTA	histidine	TTA	asparagine	CTA	aspartic acid
ATG		GTG		TTG		CTG	
ATT	STOP	GTT	glutamine	TTT	lysine	CTT	glutamic acid
ATC		GTC		TTC		CTC	
ACA	cysteine	GCA	arginine	TCA	serine	CCA	glycine
ACG		GCG		TCG		CCG	
ACT	tryptophan	GCT		TCT	arginine	CCT	
ACC		GCC		TCC		CCC	

Fig 37 The triplet codes on DNA and the amino acids they code for. Notice that there are alternative triplet codes for almost all of the 20 amino acids, as well as codes for stop.

Mutations

Mutations occur in organisms due to ***alterations in the number or structure of chromosomes***
(e.g. Down syndrome), or ***alterations to genes***.

Some mutations are detrimental to an organism, often causing death.

e.g. Mexican hairless dogs are born dead if the "hair" gene is present.

Other mutations are helpful.

e.g. Seedless grapes and oranges.

Mutations occur naturally (about 1 in 1 million sex cells have a mutation), but the rate can be increased as follows.

- Exposure to radiation from X-rays and nuclear materials.
- Exposure to chemicals such as pesticides and those in cigarettes.
- UV radiation in sunlight.

How do mutations change characteristics?

Assume a triplet sequence is as follows.

AAT	CAA	CCT	TCA
(leucine)	(valine)	(glycine)	(serine)

If the second triplet is changed to CCT, then the amino acid formed is ***glycine rather than valine***.

Hence the protein produced is quite different.

Number of Chromosomes

One chromosome can contain thousands of genes.

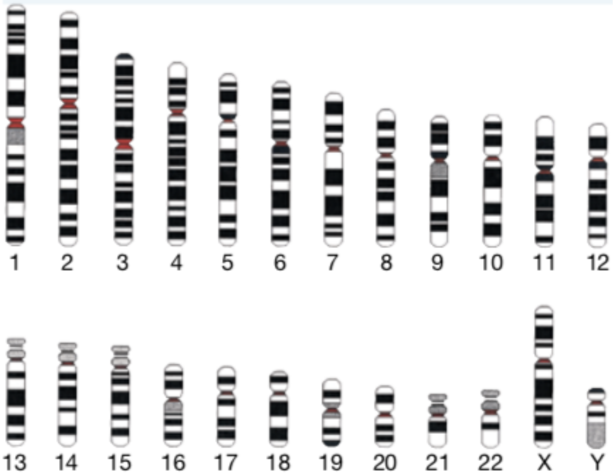
There is a characteristic number of chromosomes found in a given plant or animal species.

For humans, our normal number is 46 chromosomes (23 pairs) - this is called the ***diploid number***.

All ***body cells*** (except the reproductive cells) contain the diploid number of chromosomes.

Reproductive cells have half the normal number (23 chromosomes) - called the ***haploid number***.

Human chromosomes in order of size with banding patterns and centromere



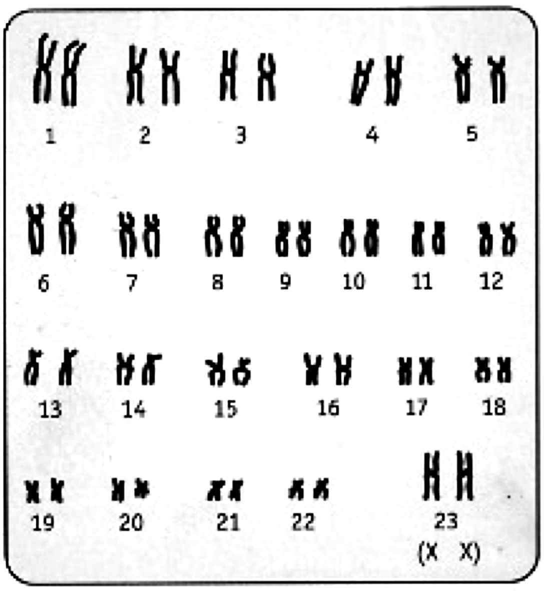


Fig 4 The 46 chromosomes in a human female are arranged in pairs based on size. The pair numbered 23 contains the sex-determining genes.

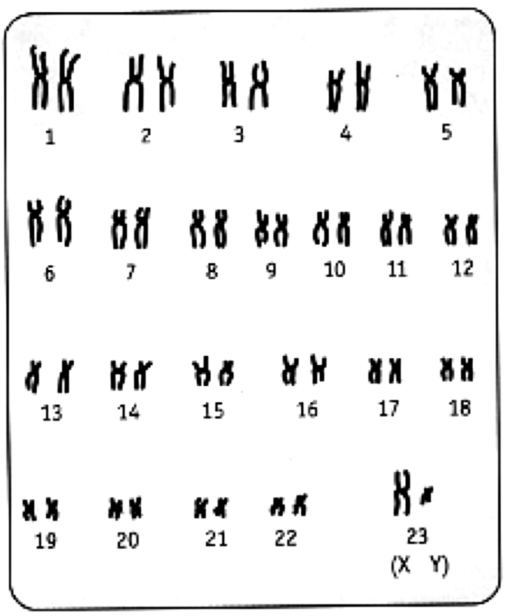


Fig 9 The chromosomes from a human male.

Chromosomes occur in pairs - one from the father, one from the mother. Each chromosome in the pair contains the same sets of genes controlling the same characteristics.

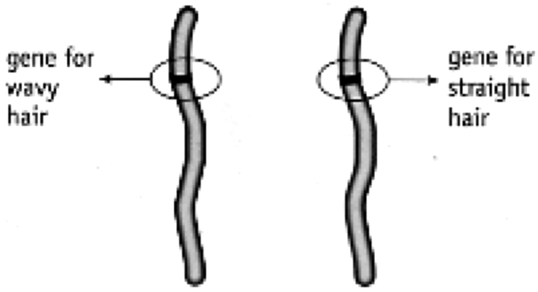
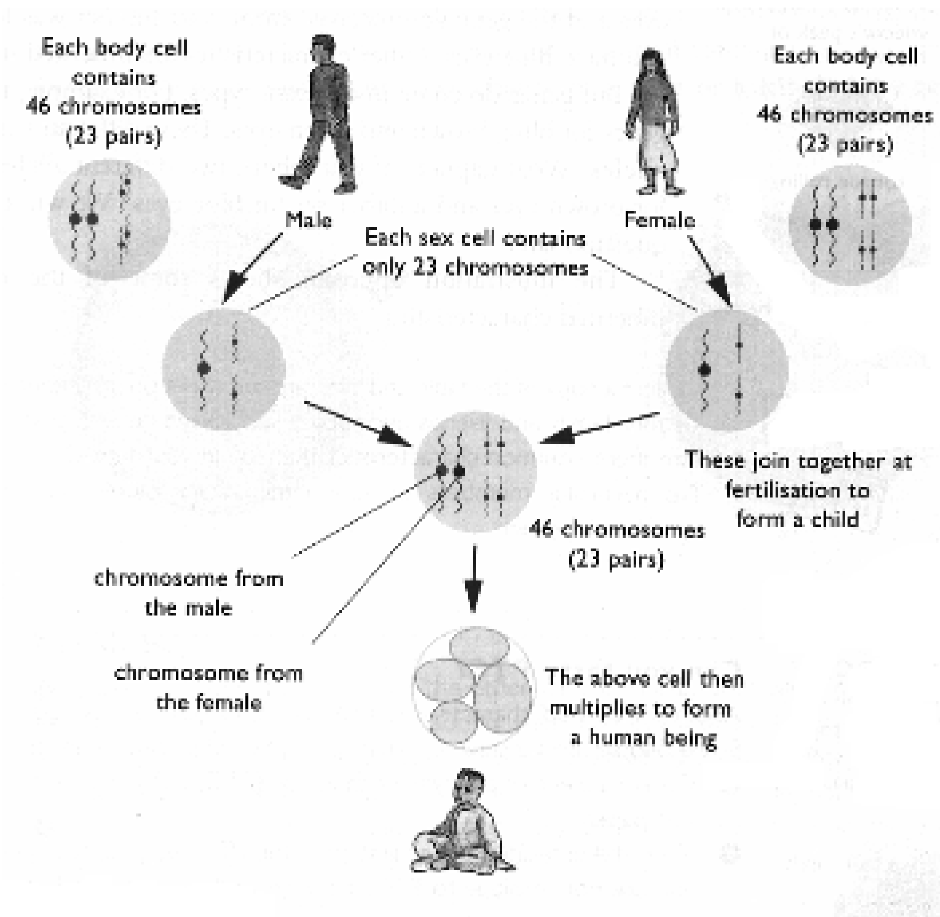


Fig 6 The genes for a particular characteristic are found at the same location on each of the chromosomes in the pair.

Another pair of chromosomes will have a different set of genes controlling other characteristics.

Mechanism Of Inheritance



The reproductive cells (called ***gametes***) from the mother and father contain 23 chromosomes each.

Upon fertilisation occurring, a single cell (called a ***zygote***) is formed with the diploid number of chromosomes - 46.

The zygote then undergoes many cell divisions to form an ***embryo*** (first 2-3 months), then a ***foetus*** (3 months till birth).

The male gamete is a ***sperm*** (***pollen*** in plants).

The female gamete is an ***egg*** or ***ovum***.